

CLASSIFICATION

CLASSIFICATION TECHNIQUES – PART III

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CLASSIFICATION TECHNIQUES

The following concepts will be introduced:

- ✓ **MULTI LAYER PERCEPTRON**
- ✓ **ARTIFICIAL NEURON**
- ✓ **INPUT/OUTPUT NEURON**
- ✓ **HIDDEN LAYER**
- ✓ **TRANSFER FUNCTION**

A **CLASSIFICATION TECHNIQUE (CLASSIFIER)** is a systematic approach to building **CLASSIFICATION MODELS** from a **DATA SET**.

CLASSIFICATION TECHNIQUES can be grouped into:

HEURISTIC; simply yet widely used like DECISION TREES, RANDOM FORESTS, NEAREST NEIGHBOR, ...

REGRESSION BASED; use a parametric conditional probability, LOGISTIC REGRESSION

SEPARATION; partition the attributes' space, SUPPORT VECTOR MACHINES and **ARTIFICIAL NEURAL NETWORKS**

PROBABILISTIC; exploit Bayes formula and compute posterior probability to classify records, NAÏVE BAYES, TREE-AUGMENTED NAÏVE BAYES, ...

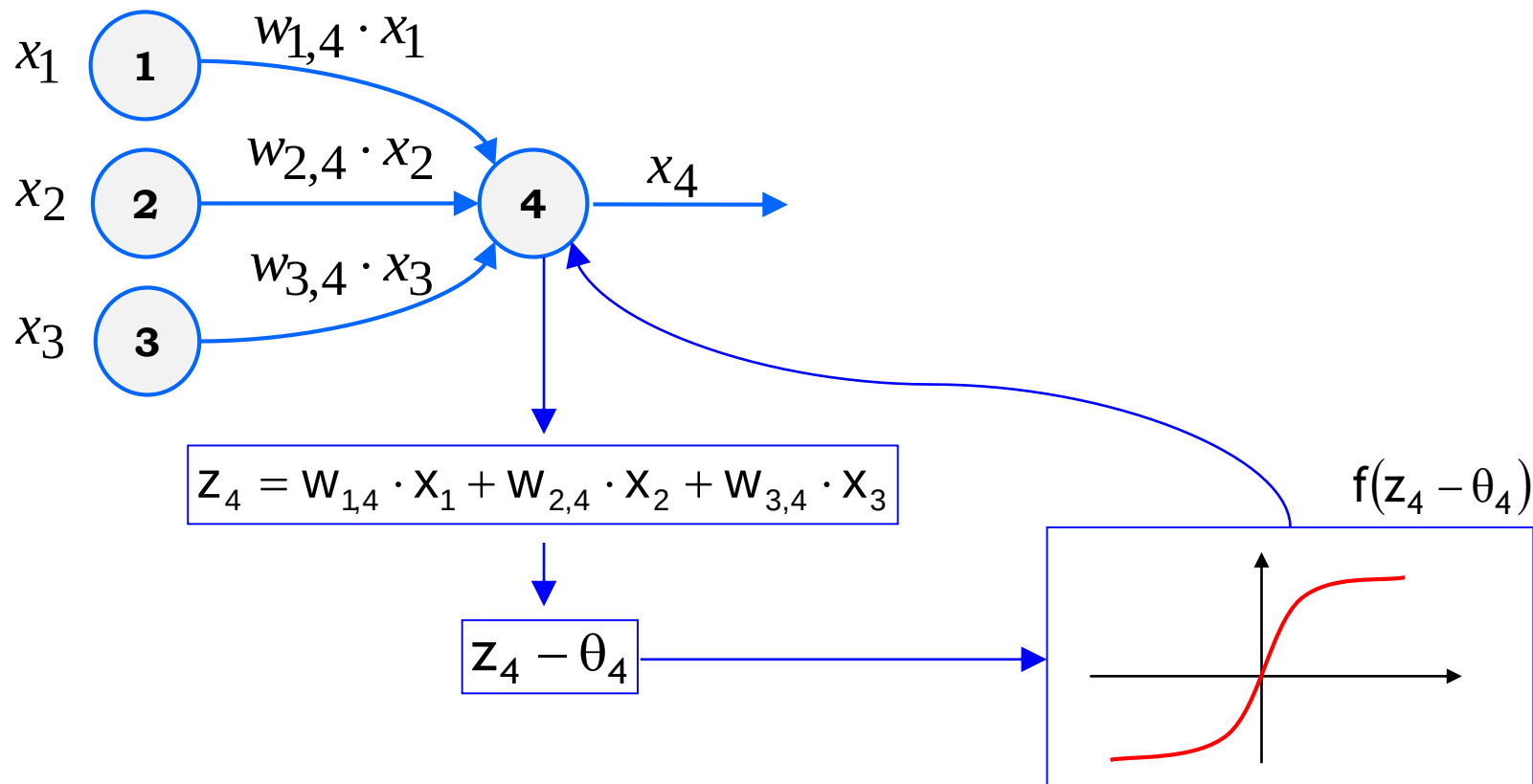
We provide a brief overview of some of this techniques even if a formal and detailed treatment of them is out of the scope of this course.

Excellent courses on these techniques are available in Machine Learning MOOC of Andrew Ng.

SEPARATION: MULTI-LAYER PERCEPTRON

2

A **MULTI-LAYER PERCEPTRON (MLP)** consists of **ARTIFICIAL NEURONS** which communicate unidirectionally, from the input attributes **X** to the Class Attribute.



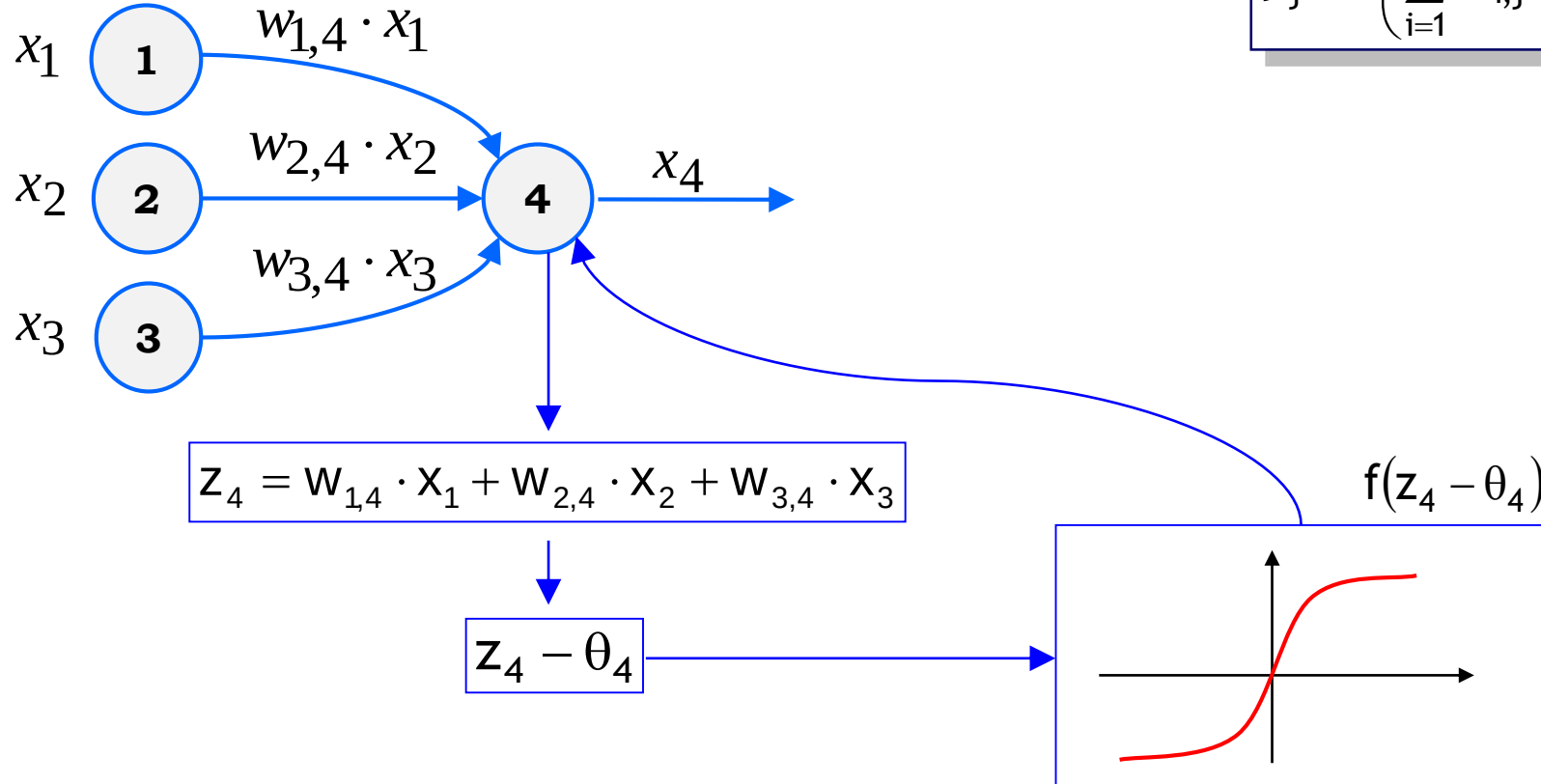
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$$y_j = f\left(\sum_{i=1}^n w_{i,j} \cdot x_i - \theta_j\right)$$



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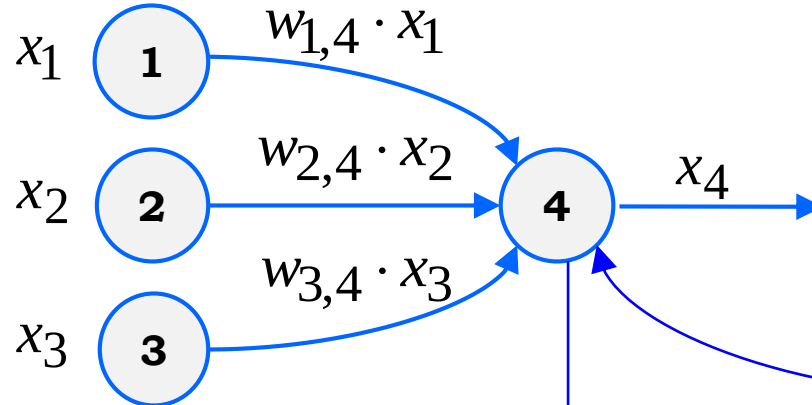
Possible **TRANSFER FUNCTIONS** are

$$f(z) = \frac{\exp(z) - \exp(-z)}{\exp(z) + \exp(-z)}$$

HYPERBOLIC TANGENT

$$f(z) = \frac{1}{1 + \exp(-z)}$$

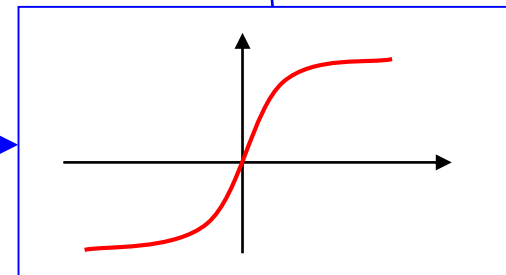
LOGISTIC



$$z_4 = w_{1,4} \cdot x_1 + w_{2,4} \cdot x_2 + w_{3,4} \cdot x_3$$

$$z_4 - \theta_4$$

$$f(z_4 - \theta_4)$$



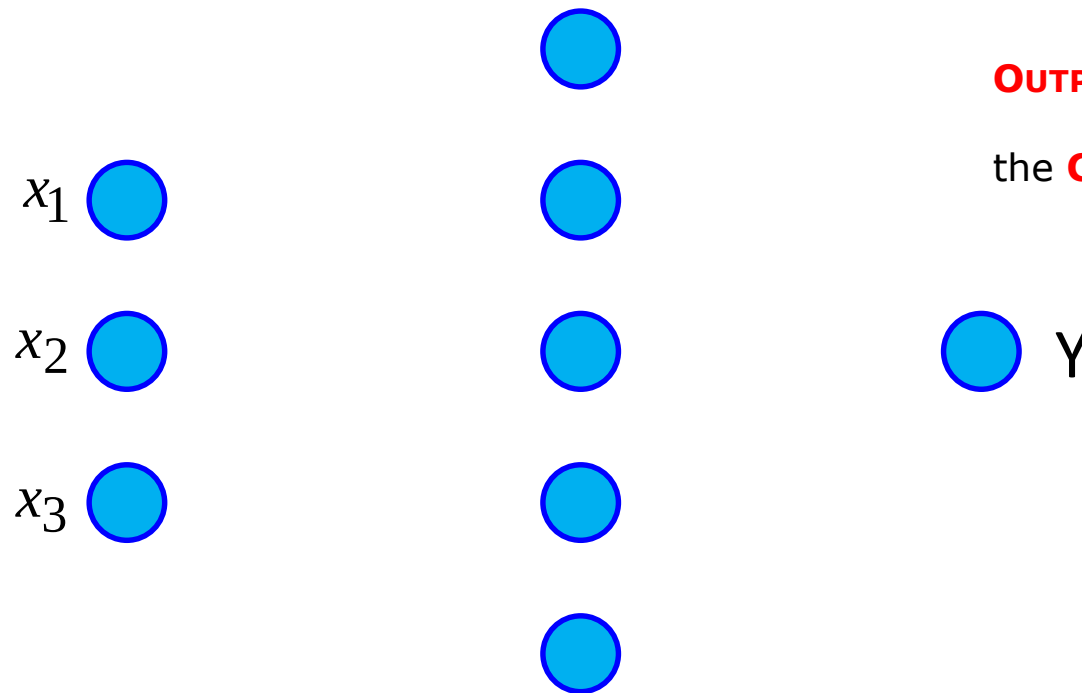
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A **MULTI-LAYER PERCEPTRON (MLP)** consists of **INPUT NEURONS**, **HIDDEN NEURONS** and an **OUTPUT NEURON**.

INPUT NEURONS are associated with the **EXPLANATORY ATTRIBUTES**.

OUTPUT NEURON is associated with the **CLASS ATTRIBUTE**.



INPUT NEURONS

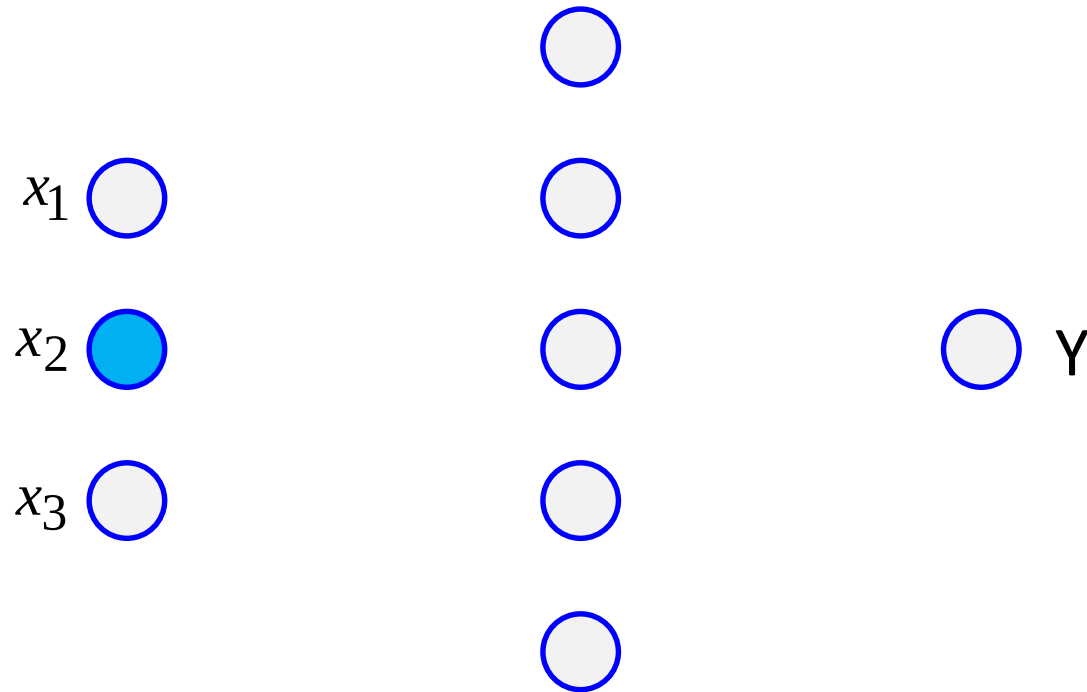
HIDDEN NEURONS

OUTPUT NEURON

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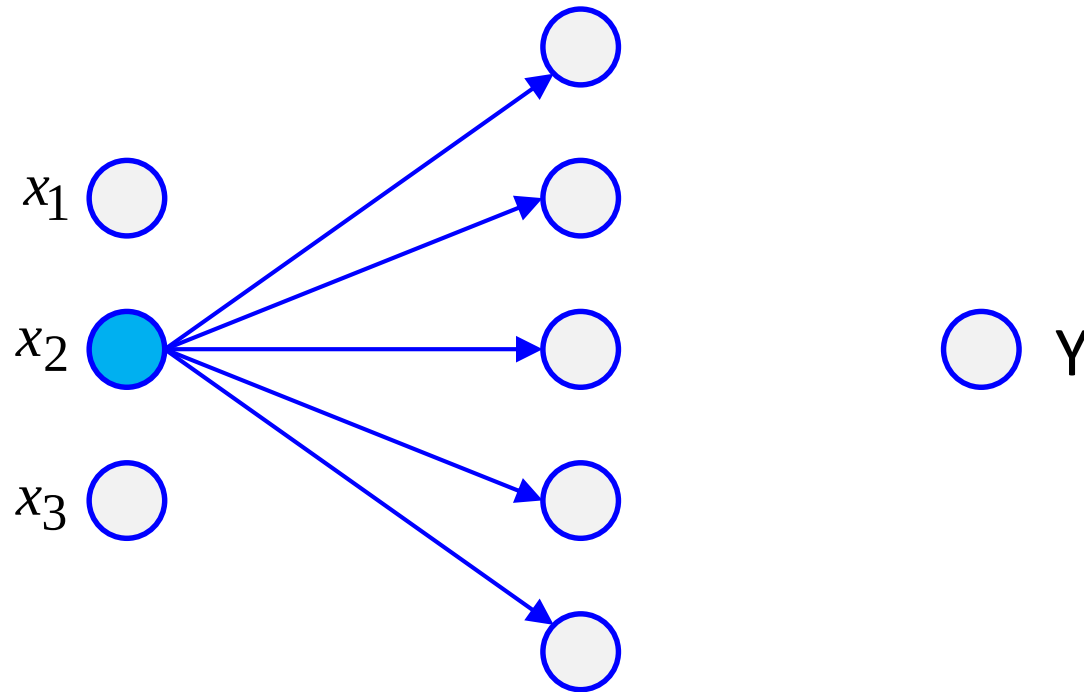
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Each **INPUT NEURON** is connected through directional links with all **HIDDEN NEURONS**.



INPUT NEURONS

HIDDEN NEURONS

OUTPUT NEURON

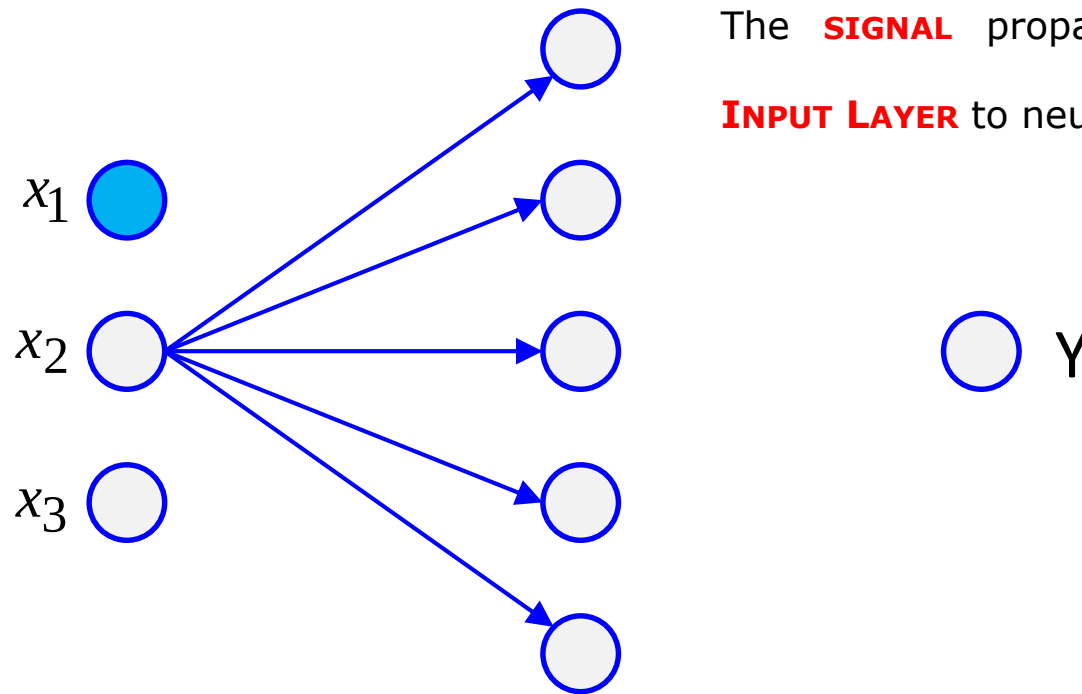
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The **SIGNAL** propagates from neurons of the **INPUT LAYER** to neurons of the **HIDDEN LAYER**.



INPUT NEURONS

HIDDEN NEURONS

OUTPUT NEURON

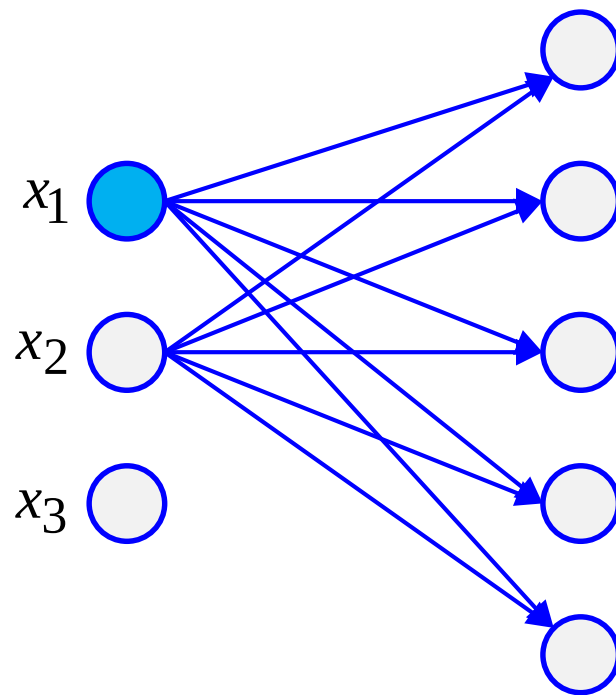
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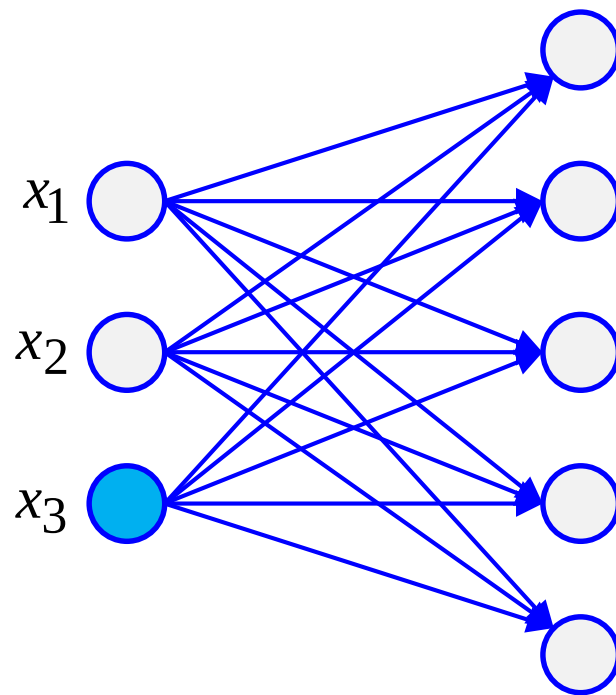
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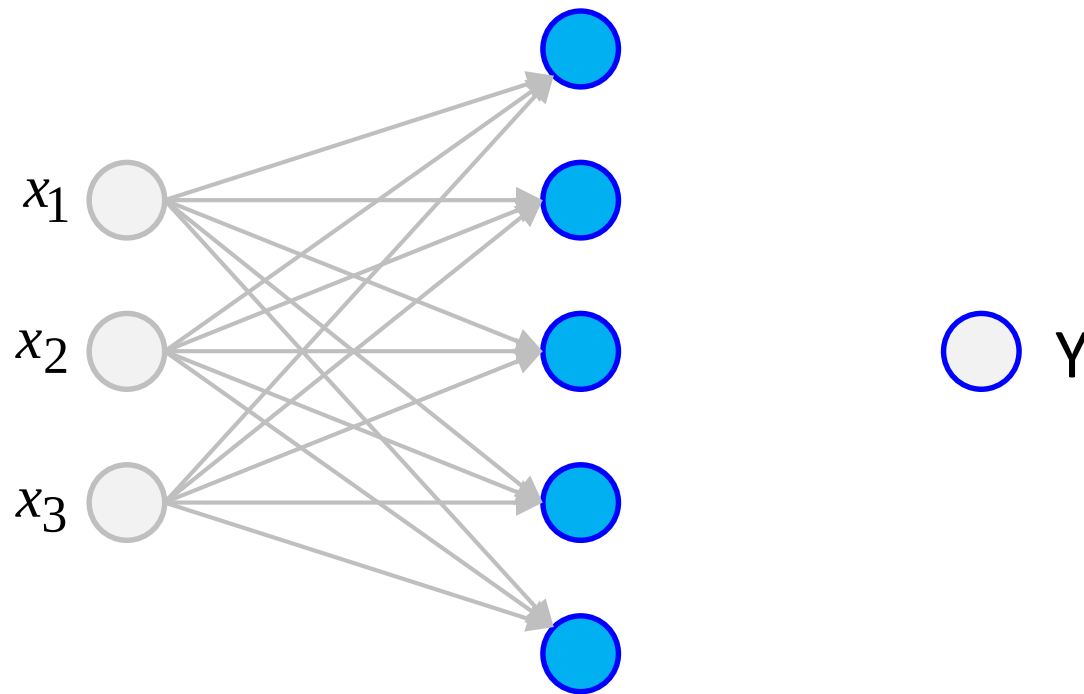
HIDDEN NEURONS

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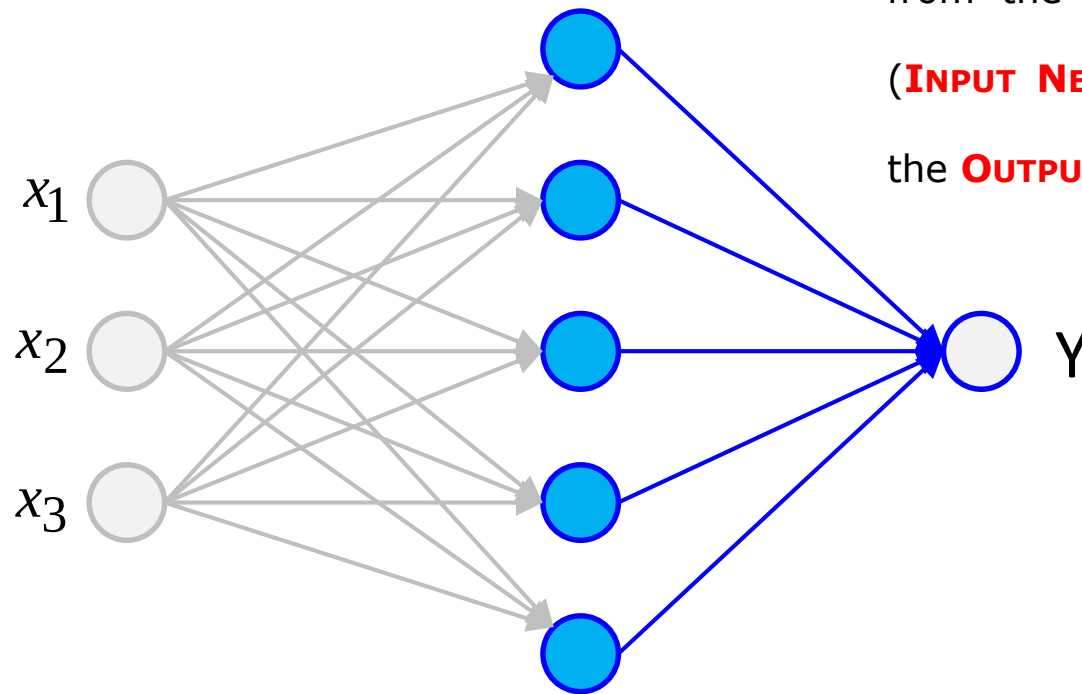
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Once all neurons of the **HIDDEN LAYER** (**HIDDEN NEURONS**) received the **SIGNAL** from the neurons of the **INPUT LAYER** (**INPUT NEURONS**) the **SIGNAL** is sent to the **OUTPUT NEURON**.



INPUT NEURONS

HIDDEN NEURONS

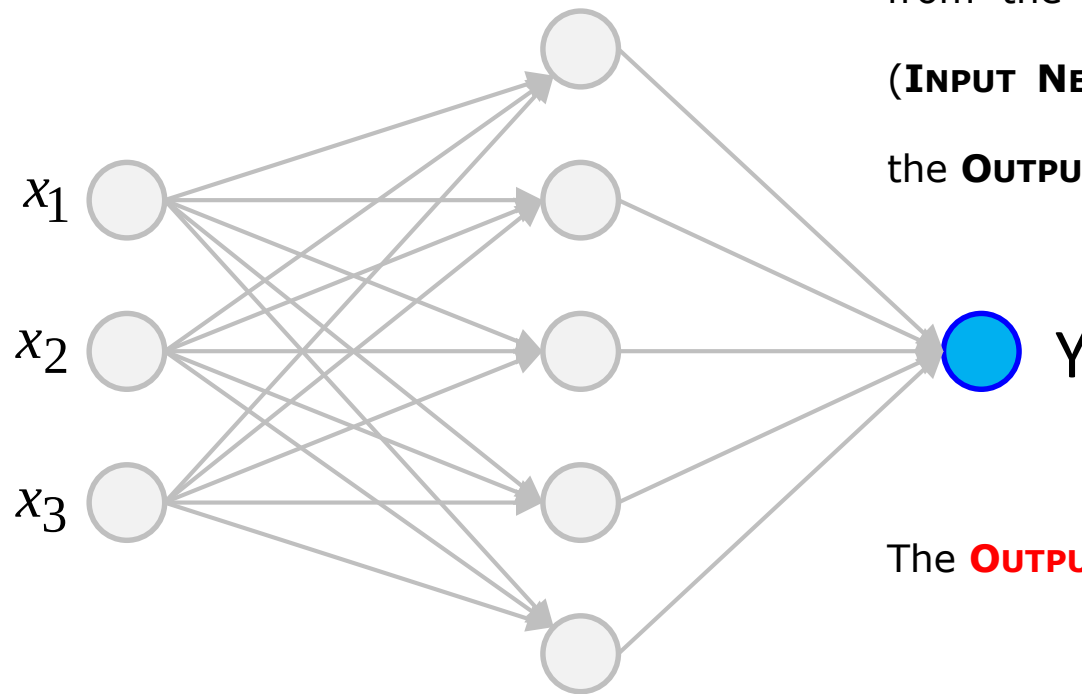
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Once all neurons of the **HIDDEN LAYER** (**HIDDEN NEURONS**) received the **SIGNAL** from the neurons of the **INPUT LAYER** (**INPUT NEURONS**) the **SIGNAL** is sent to the **OUTPUT NEURON**.



The **OUTPUT NEURON** labels the record.

INPUT NEURONS

HIDDEN NEURONS

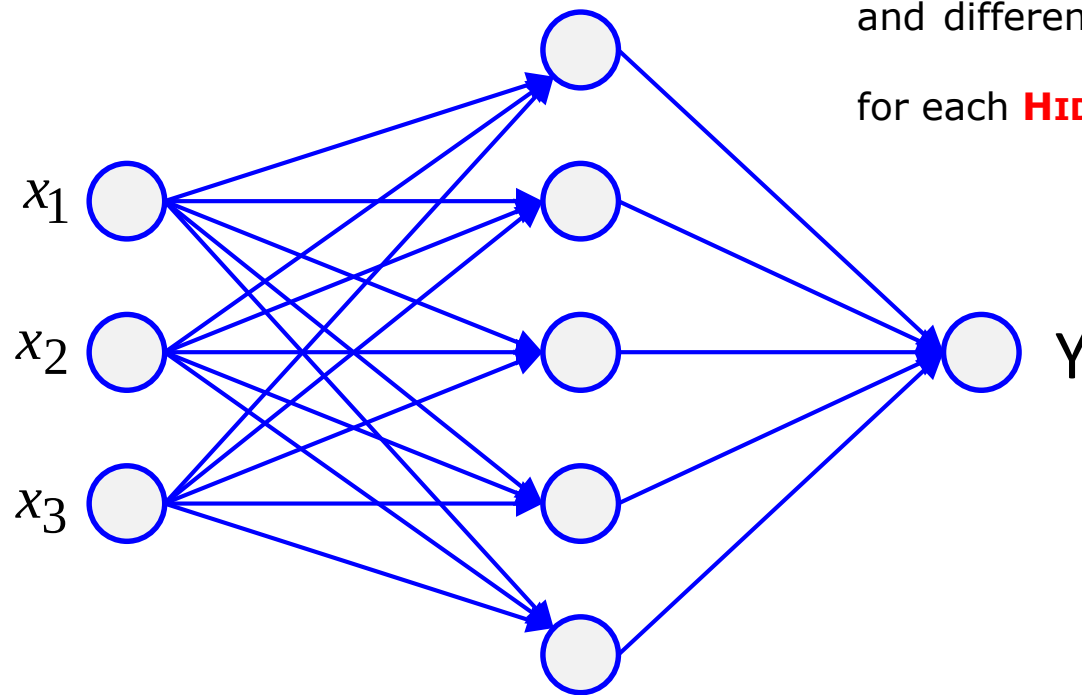
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A **MULTI-LAYER PERCEPTRON (MLP)** consists of **INPUT NEURONS**, **HIDDEN NEURONS** and an **OUTPUT NEURON**.

MLPs can have **DIFFERENT ARCHITECTURES**,
i.e., different number of **HIDDEN LAYERS**
and different number of **HIDDEN NEURONS**
for each **HIDDEN LAYER**.



INPUT NEURONS

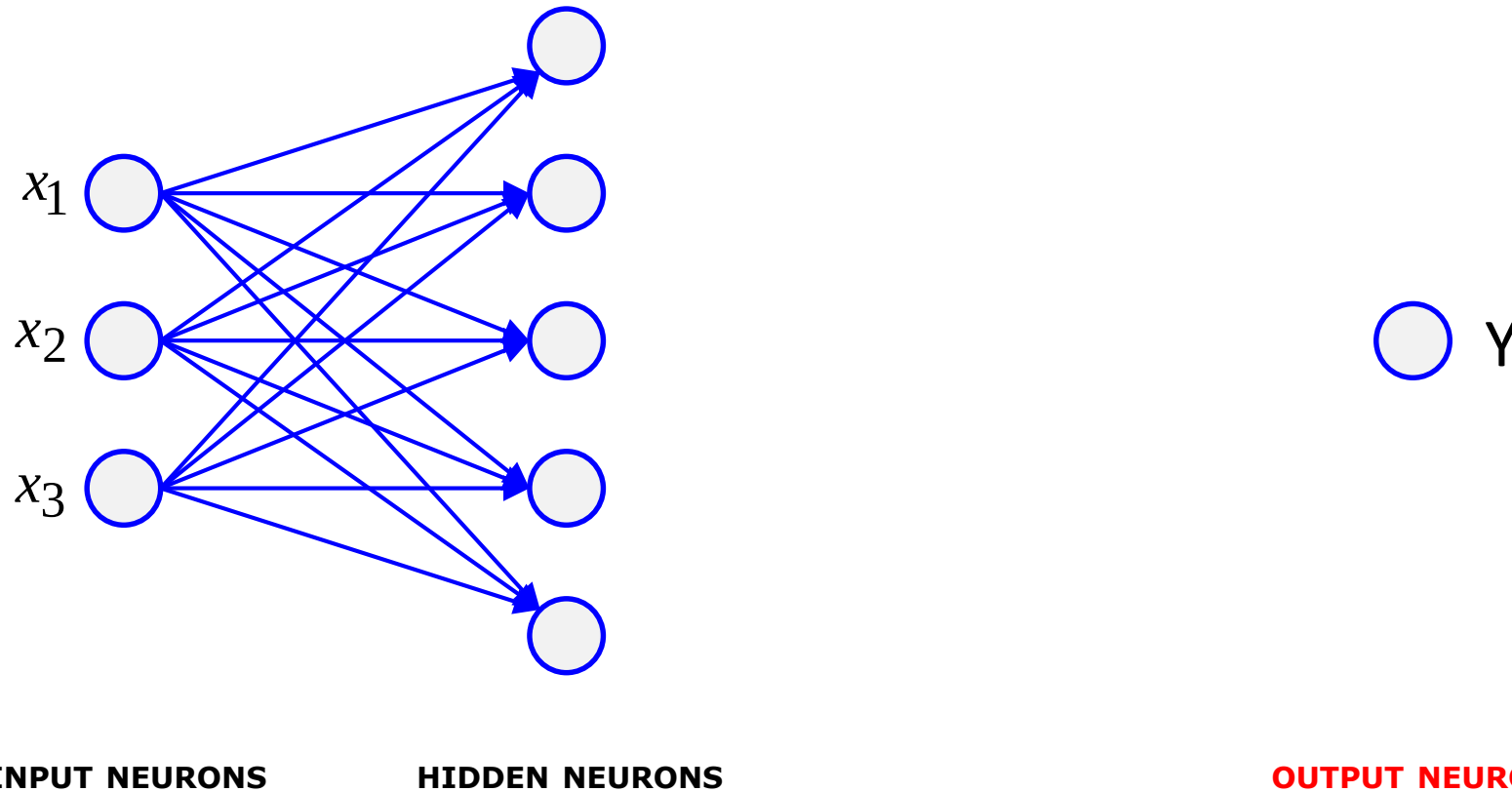
HIDDEN NEURONS

OUTPUT NEURON

SEPARATION: MULTI-LAYER PERCEPTRON

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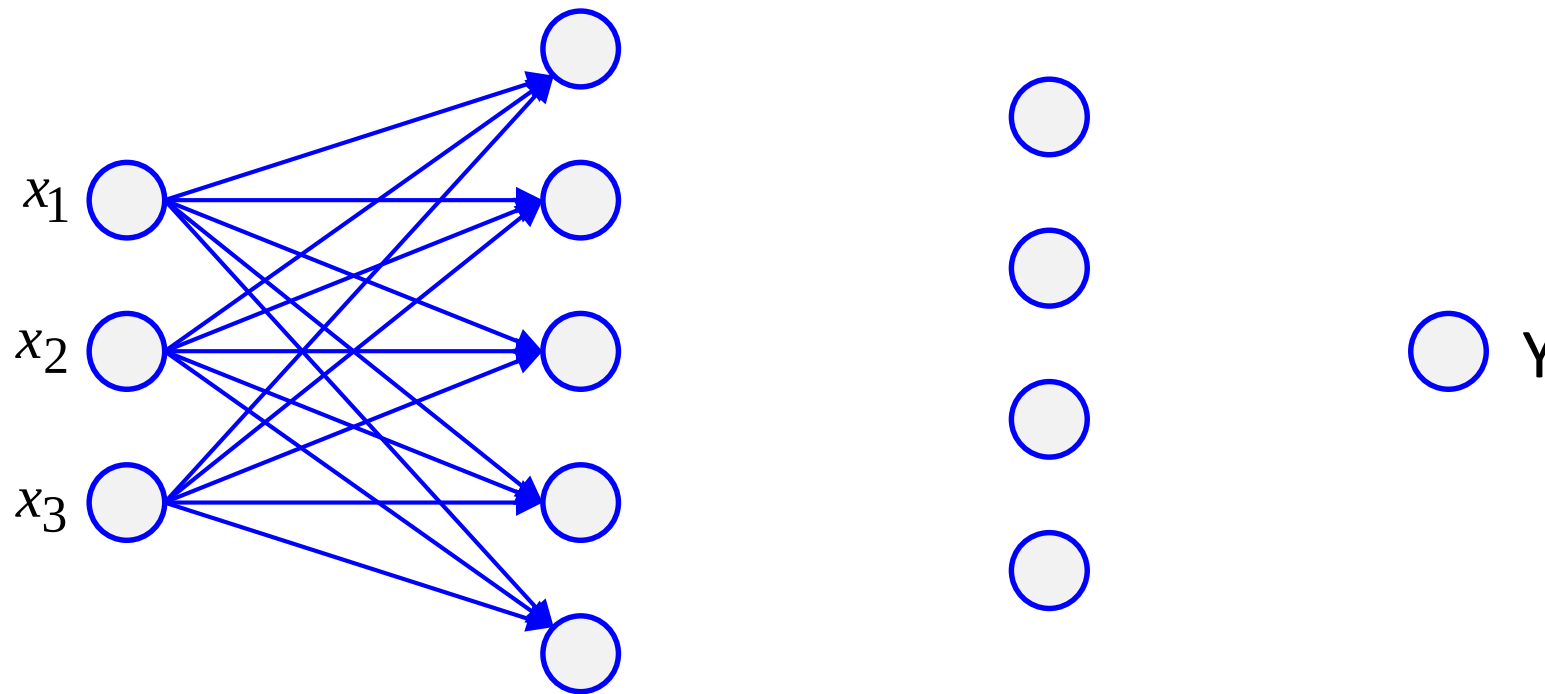
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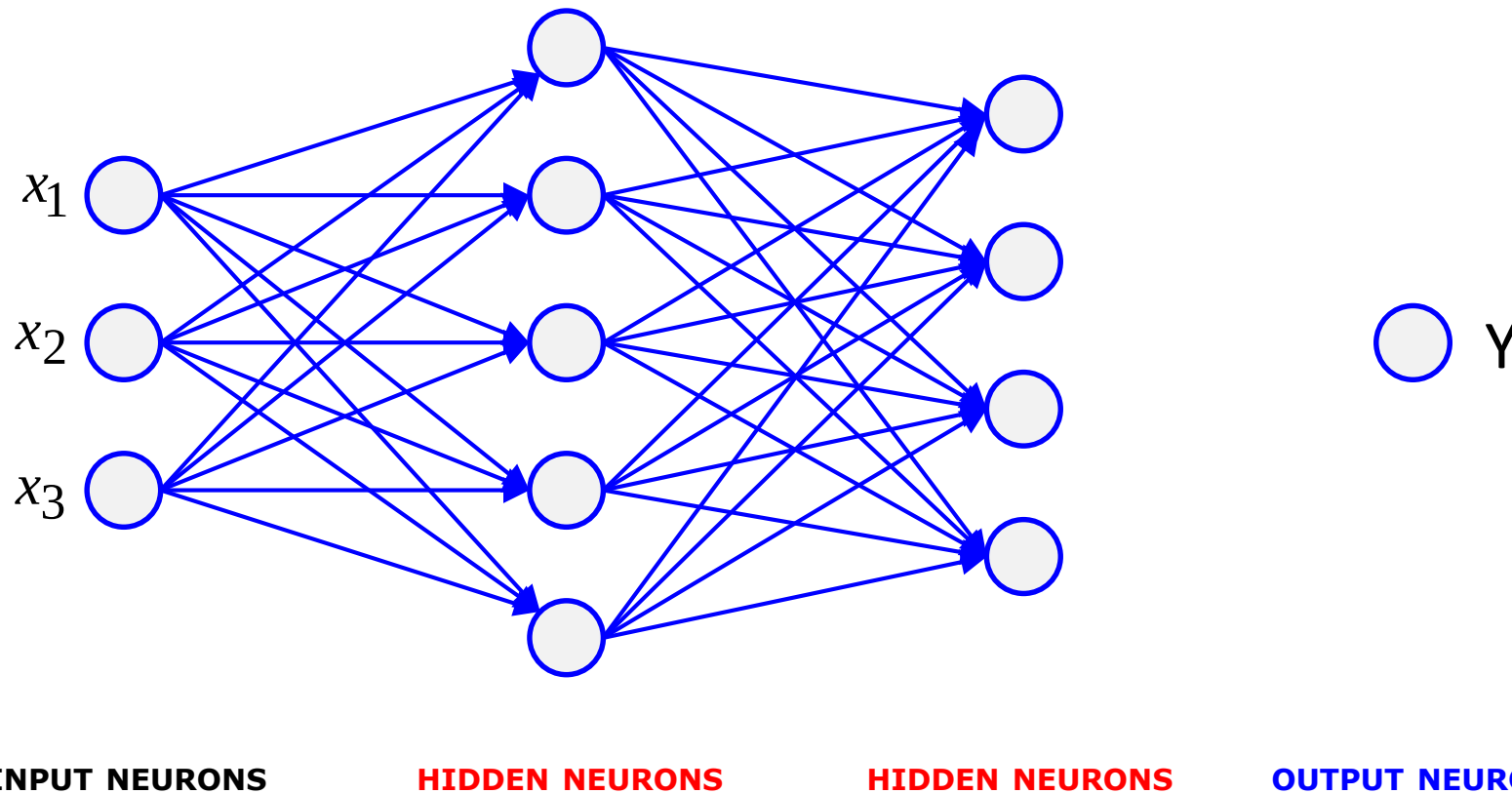
HIDDEN NEURONS

OUTPUT NEURON

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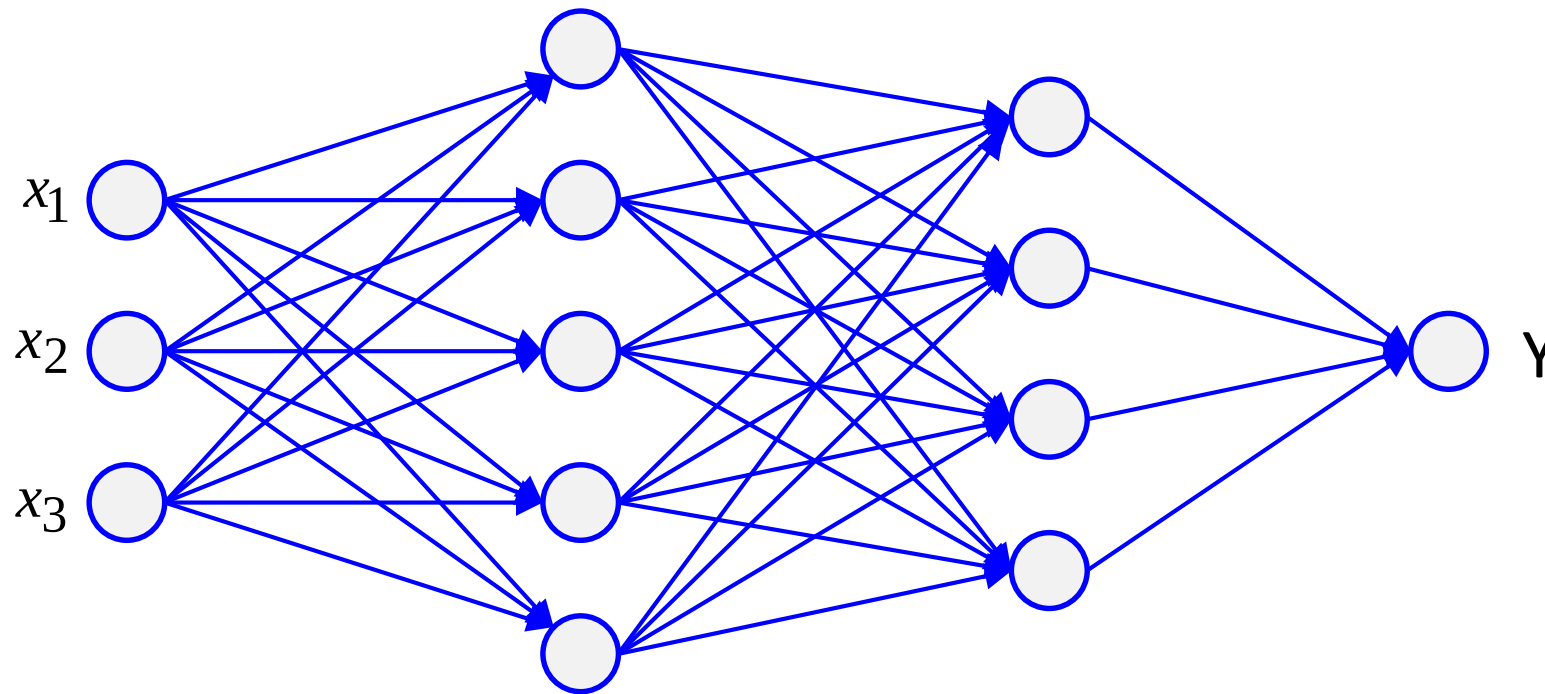
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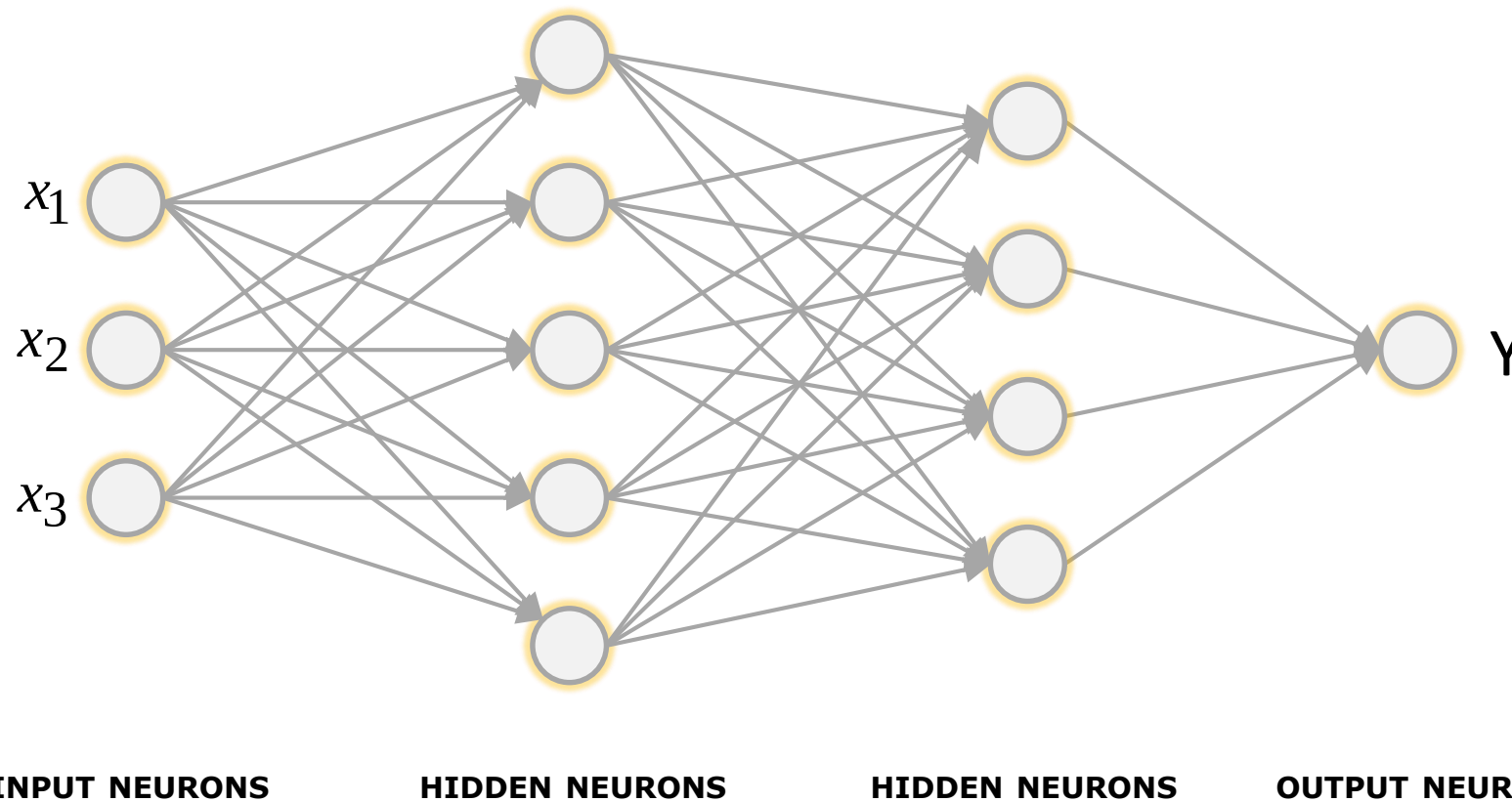
HIDDEN NEURONS

HIDDEN NEURONS

OUTPUT NEURON

The main **DESIGN ISSUES** are

- **HOW MANY HIDDEN LAYERS?**
- **HOW MANY HIDDEN NEURONS FOR HIDDEN LAYER?**



MLP LEARNING IS A DIFFICULT TASK, different **OPTIMIZATION MEASURES** have been proposed in the specialized literature, **MANY DIFFERENT LEARNING ALGORITHMS** have been designed and described but **MULTIPLE LOCAL MINIMA** usually exist and thus it is always extremely difficult to learn an MLP.

